MILITARY SPECIFICATION

FIN XM25

1. SCOPE

1.1 This specification covers Fin Assembly XM25.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Munitions Command

F11751154	- Fin, XM25
LI11751154	- Index of Inspection Equipment List for Fin XM25

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, New York.)

ASTM Method - American Society for Testing and Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Mechanical properties.-Mechanical properties shall be in accordance with the requirements of the applicable drawings and specification.

3.3 Workmanship.-The requirements for workmanship are as shown on the applicable drawing, referenced specifications and the following:

3.3.1 Metal defective.-Soundness of the material shall be as required by the applicable material specifications. Extrusion stock used in manufacture of fins shall be free from cracks, laminations, porosity and pipe type defects.

3.3.2 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawings.

3.3.3 Burr.-No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.3.4 Foreign matter.-All parts shall be free of chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities

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4.1 (Cont'd)

suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define Quality Assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-25500 except as specified herein.

4.2 First Article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 10 fins. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used in regular production and shall be obtained from the same source of supply. A first article or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, procedures, material used, drawings, specifications or source of supply such as to significantly affect product uniformity as determined by the Government.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.2 Extrusion lots.-A lot shall consist of not more than 5,000 fins from extruded stock made by the same process, of the same alloy and temper which has been aged in the same furnace charge.

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4.3.2 Examination.-One-hundred percent examination shall be performed for all critical defects. Examination for Major and Minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination. Four power (4X) magnification may be used in performing visual examination.

4.3.2.1 Control tool manufacture.-For dimensions marked control tool manufacture, select and gage one fin from each machine in operation every half hour or fraction thereof, and whenever tooling is changed. Prior to any tooling change the last fin produced from the old tooling shall be gaged for the requirements of the applicable drawing. All material produced prior to periodic inspection shall be rejected and screened whenever a defective characteristic is found and corrective action must be taken.

4.3.2.2 Fin, Drawing F11751154.

Defect	Method of Inspection
CRITICAL	
1. Flash hole(s) missing or excessive in number	Visual
2. Metal defective	Visual
MAJOR	
101. Runout of outside diameter of boom with pitch diameter of body thread and front face	Gage
102. Runout of pitch diameter of primer thread with pitch diameter of body thread and front and rear faces	Gage
103. Runout of outside diameter of fin blades with pitch diameter of body thread and front face	Gage
104. Straightness of fin blades after canting blades (horizontal)	Gage
105. Straightness of fin blades after canting blades (vertical)	Gage
106. True position of primer cavity with pitch diameter of primer thread for min. length	Gage
107. Pitch diameter of body thread, max.	Gage
108. Pitch diameter of body thread, min.	Gage
109. Minor diameter of body thread, max.	Gage
110. Pitch diameter of primer thread, max.	Gage
111. Pitch diameter of primer thread, min.	Gage
112. Minor diameter of primer thread, max.	Gage

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4.3.2.2 (Cont'd)

	Defect	Method of Inspection
113.	Diameter of primer cavity	Gage
114.	Diameter of boom	Gage
115.	Diameter of fin blade counterbore at rear of fin	Gage
116.	Diameter of fin blades (large)	Gage
117.	Diameter of fin blades (small)	Gage
118.	Depth of primer cavity	Gage
119.	Depth to bottom of body thread cavity	Gage
120.	Depth from rear face of fin blades to rear face of fin hub	Gage
121.	Length of body thread, min.	Gage
122.	Length of primer thread, min.	Gage
123.	Length of flat at largest diameter of fin blades	Gage
124.	Length from rear of fin blade to start of fin blade	Gage
125.	Length, total	Gage
126.	Location of datum on forward edge of fin blade	Gage
127.	Height of cant, fin blade	Gage
128.	Fin blade(s) distorted or damaged	Visual
129.	Fin blade(s) canted in improper direction	Visual
130.	Thread(s) damaged or not full	Visual
MINOR		
201.	Diameter of hub between fin blades	Gage
202.	Diameter of body thread counterbore	Gage
203.	Depth of body thread counterbore	Gage
204.	Depth of primer thread max.	Gage
205.	Location of largest outside fin diameter flat from rear of fin	Gage
206.	Angle between fin blades	Gage
207.	Thickness of fin blades	Gage
208.	Radii or chamfers missing or incorrect	Visual (see 6.1)
209.	Surface finish improper	Visual (see 6.1)
210.	Burr	Visual
211.	Foreign matter	Visual
212.	Marking missing, incorrect or illegible	Visual

4.3.2.3 Packed Carton prior to sealing.

CRITICAL

None defined

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4.3.2.3 (Cont'd)

Defect	Method of Inspection
MAJOR	
None defined	
MINOR	
201. Support(s) missing or damaged	Visual
202. Divider(s) missing or damaged	Visual
203. Improperly packed	Visual

4.3.2.4 Packed Carton after sealing.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Cartons damaged (cut or damaged through all layers)	Visual
203. Carton not properly sealed	Visual
204. Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with provisions of 4.4.

4.3.3.1 Mechanical properties.

4.3.3.1.2 Tension test.

4.3.3.1.2.1 Extrusion lots.-From the extrusion lot (as defined by 4.3.1.2) two fins shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirement of the applicable drawing shall be cause for rejection of the lot.

4.4 Test methods and procedures.

4.4.1 Mechanical properties.

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4.4.1.2 Tension.

4.4.1.2.1 Fins.-From the sample selected in accordance with 4.3.3.1.2.1, one tension test specimen conforming to the standard "F" type specimen shall be taken longitudinally from a fin blade and one tension test specimen conforming to the largest obtainable standard "R" type specimen shall be taken longitudinally from the boom section. The number of specimens and their location is to determine the acceptability of the mechanical properties requirements and with the approval of the procuring agency, the Government shall not be restricted from testing additional longitudinal specimens at other locations to determine that the mechanical properties meet the drawing requirements. The criteria for performing the tests shall be as prescribed by ASTM Method E-8.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.-Packaging, packing and marking shall be in accordance with the requirements of Level C in MIL-C-1169.

5.2 Data card.-Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.3 Ordering data.-Procurement documents should specify the title, number and date of this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1310-A216

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